Quantigration RMA | Data Analysis Report

Project Two | Educational SQL analysis
Findings reflect the DAD 220 larger Codio lab dataset, not the public synthetic demonstration dataset.

Purpose and scope

The product manager requested a summary of where product returns are associated with customers and which product descriptions account for the most RMA records. Using SQL joins and grouped counts, the analysis examined 37,566 RMA rows.

Returns by state

The SQL query returned 48 state categories. Massachusetts had the most RMAs (972), followed by Arkansas (844) and Oregon (840). South Carolina had the fewest (702). The gap between the highest and lowest is 270 RMA records. These are absolute counts, so states with higher sales volume could naturally have more returns.

Returns by product description

Product description	RMA rows	Share
Basic Switch 10/100/1000 BaseT 48 port	8,282	22.05%
Enterprise Switch 40GigE SFP+ 48 port	6,118	16.29%
Basic Switch 10/100/1000 BaseT 24 port	33	0.09%

Nine product descriptions appeared in the query. The two largest categories together made up 38.34% of all RMA records. This highlights which products contribute the largest number of return records, but it does not establish which has the highest return rate relative to sales.

Data checks and caveats

Both the total-RMA query and the order-join count returned 37,566, supporting complete linkage from RMA records to orders in this dataset. These checks do not establish that RMA entries are free of duplicates, have accurate reasons, reflect completed returns, or join to valid customers. Product descriptions and state values may also contain quality issues not evaluated by these queries.

Recommendations

Prioritize review of return reasons and statuses for the largest product categories; compare state and product return counts against corresponding order or units-sold totals; and track trends over time if dates are available. Avoid inferring product defects or unusual state behavior from raw return counts alone.

Method

SQL used `RMA` joined to `Orders` and, for state summaries, to `Customers`; `COUNT`, `GROUP BY`, and `ORDER BY` produced counts. Each product’s share is its RMA count divided by total RMA rows, rounded to two decimals. SQL and screenshots are provided in this folder.